

REGHURAM KESAVAN

AERODYNAMICS · CFD · AIRCRAFT PERFORMANCE · PYTHON · FLIGHT PHYSICS

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PROFESSIONAL PROFILE

Mechanical Engineering graduate completing the Master in Aerospace Engineering at ISAE-SUPAERO, specialising in Advanced Aerodynamics & Propulsion. Hands-on technical background spans low-speed wind-tunnel testing, Transition-SST RANS high-lift analysis, Python low-order aerodynamic modelling (3D VLM/Trefftz drag), aircraft configuration trades, and propulsion–airframe interaction, alongside industrial aerospace structural test and mechanical design experience at Lilium. Actively targeting aircraft aerodynamics, flight physics, and scientific aerodynamic tool development.

AERODYNAMICS & FLIGHT PHYSICS PROJECTS

High-Lift Aerodynamics: Wind-Tunnel Testing & Transition-SST RANS ISAE-SUPAERO | Oct – Dec 2025

- **Experimental Tunnel Testing:** Executed low-speed wind-tunnel testing on the HLHSD4 high-lift airfoil at $Re = 1.0 \times 10^6$ across multiple angles of attack to establish baseline lift, drag, and stall boundary behavior.
- **RANS & Grid Convergence:** Generated structured multi-block grids ($y^+ < 1$; 11k, 23k, 40k cells) for Transition-SST RANS in ANSYS Fluent; achieved ~0.15% CL and ~2% CD medium-to-fine convergence at 10° and correlated computed suction-side pressure and skin-friction trends against XFOIL to interpret separation progression.

VOROM-X: Aircraft Aerodynamic Design & Flight-Physics Prototype Python / Research | Jan 2024 – Present

- **Low-Fidelity VLM & Trefftz Drag:** Developed an object-oriented Python 3D Vortex Lattice Method (VLM) tool incorporating Trefftz-plane induced drag integration for rapid wing planform trades, spanwise circulation/loading analysis, and lift-to-drag optimization.
- **Stability, Trim & Model Limits:** Automated longitudinal trim, neutral point, and static stability derivative calculation for swept/twisted configurations; benchmarked aerodynamic polar trends against published experimental data and XFLR5 to rigorously define low-order operational limits.

Aerodynamics of Passenger Aircraft with Distributed Electric Propulsion (DEP) B.Tech Thesis | Aug 2021 – May 2022

- **Propulsion–Airframe Interaction:** Investigated aerodynamic coupling between a transport wing and simplified stationary nacelle installations using steady RANS in ANSYS Fluent within a 4-engineer team, evaluating flow acceleration and local surface pressure fields.
- **Boundary Layer & Wake Analysis:** Quantified chordwise velocity profiles and downstream boundary-layer changes across baseline vs. installed configurations to evaluate regional lift-to-drag trade-offs and nacelle integration effects.

Scientific Workflow Automation & Methods (ApexSim · TRACE FanGEO) Python / Research | 2024 – Present

- **CFD Automation & Surrogate Databases (ApexSim):** Engineered Python pipelines for automated SU2 RANS execution and grid-convergence tracking; enforced numerical evidence gates (boundary remediation: $\Delta CL: 7.22\% \rightarrow 1.52\%$, $\Delta CD: 4.21\% \rightarrow 2.93\%$) and fitted Kriging surrogates to generate continuous aero performance databases.
- **Turbomachinery Design Workflow (TRACE FanGEO):** Developed a requirement-driven Python fan throughflow and 3D blade geometry preprocessor across 41 spanwise sections at ISAE-SUPAERO; automated velocity-triangle closure, stagger/camber parameterisation, and interface consistency checks.

PROFESSIONAL AEROSPACE EXPERIENCE

Feynman Aerospace LLP — Aircraft Design Intern Mumbai, India | Apr – Oct 2021

- **Aerodynamic & Configuration Analysis:** Performed low-Reynolds-number aerodynamic and configuration analysis for an ~2 m fixed-wing UAV using XFLR5 and ANSYS Fluent; generated 3D surface geometry in SolidWorks and supported prototype flight testing and subsequent aero design iteration.

Lilium eAircraft GmbH — Mechanical Design Engineer Intern Munich, Germany | Apr – Nov 2024

- **Test Hardware & Cross-Discipline Design:** Defined structural test requirements and designed custom test fixtures and support hardware for lightweight composite sandwich panels in Siemens NX with ASME Y14.5 GD&T, ensuring seamless interface alignment with physical test benches.
- **Pre-Test FEA & Configuration Management:** Conducted pre-test nonlinear FEM in ANSYS Workbench to predict stress concentrations and fixture stiffness, managing multi-part assemblies, drawing packages, and engineering changes under controlled PDM workflows in Teamcenter.

EDUCATION

ISAE-SUPAERO — Master in Aerospace Engineering Toulouse, France | 2025 – Expected Dec 2026

- **Major:** Advanced Aerodynamics & Propulsion. **Key Coursework & Verified Grades:** Advanced Aerodynamics (15/20), Numerical Fluid Mechanics (15/20), Fluid–Structure Interaction (16/20), Project Management (20/20), Computational Aeroacoustics.

Technical University of Munich (TUM) — Graduate Aerospace Engineering Studies Munich, Germany | 2023 – 2025

- **Completed 57 ECTS in:** Aircraft Design, Aircraft Performance, Theoretical & Experimental Aerodynamics, Composite Materials & Structures, Control Engineering, Aeroelasticity, Aerospace Structures.

APJ Abdul Kalam Technological University — B.Tech (Honours), Mechanical Engineering Kerala, India | 2018 – 2022

- **Official CGPA:** 9.28 / 10.0. Core modules: Fluid Mechanics, Gas Dynamics, Heat Transfer, Machine Design, Finite Element Analysis.

TECHNICAL SKILLS

- **Aerodynamics & Flight Physics:** RANS CFD, High-Lift Aerodynamics, Aircraft Performance & Trim, Static Margin & Stability Derivatives, Low-Order Aerodynamic Modelling (3D VLM/Trefftz), Propulsion–Airframe Interaction
- **CFD & Aero Tools:** ANSYS Fluent, SU2, XFOIL, XFLR5, Structured Multi-Block Meshing, Boundary Layer Inflation ($y^+ < 1$), Grid Convergence (GCI)
- **Programming & Computing:** Python (NumPy, SciPy, Matplotlib), MATLAB, Git / GitHub, Linux/Bash, Workflow Automation, Parametric Scripting
- **Design, FEA & PLM:** Siemens NX, SolidWorks, GD&T (ASME Y14.5), ANSYS Workbench (Nonlinear/Static Structural FEA), Teamcenter
- **Methods & Testing:** Wind-Tunnel Test Execution, Surrogate Modelling, Design of Experiments (DOE), Aerodynamic Database Assembly, Numerical Verification